



Machine learning in project schedule creation: a systematic literature review

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Abstract

Addressing the growing demand for autonomous project scheduling, this study explores the use of machine learning (ML) for project schedule creation. It is a well-known fact that ML surpasses human capabilities in analyzing extensive data. For this reason, ML is a promising domain for organizations that gather project experience. However, the current ML adoption in project management remains limited, lacking systematic studies demonstrating its potential. This is the first study that presents a comprehensive review of the current state of knowledge in the project schedule creation area. We summarize the ML techniques used in each schedule creation area and investigate their demonstrated potential in improving schedule creation processes. Moreover, we analyze in which industries such methods are most often used. Furthermore, we explore anticipated advancements, highlight potential benefits, and address current limitations of ML applications in the domain of schedule creation. We also uncover key gaps in research, discuss related challenges, and emerging trends. Additionally, we elaborate on practical implementation strategies for ML-based schedule creation, emphasizing that while ML has the potential to enhance project scheduling, full automation is unlikely in the near future. Instead, automating specific subprocesses appears to be the most practical approach for integrating ML into schedule creation at this stage.

Keywords Machine learning · Intelligent systems · Schedule creation · Project schedule management · Artificial intelligence

1 Introduction

To set the stage, we begin with a brief discussion of artificial intelligence (AI), its evolution, and its expansion into machine learning (ML). We also explore the role of AI and ML in project scheduling, followed by a look at current challenges and potential. The introduction also outlines key concepts and their relevance to project scheduling. The distinct domains of AI, ML, and scheduling converge in this paper, with a focus on the project schedule management area.

The inception of artificial intelligence (AI) dates back to before the 1950s, when Alan Turing elaborated on thinking machines and documented his ideas (Turing, 1950). For the purpose of this paper, we follow a practical definition of

the artificial intelligence concept coined by John McCarthy (2007), which states that “it is science and engineering of making intelligent machines, especially intelligent computer programs” (p. 2). The AI field aims to make computers perceive, understand, and act in a way that is still only possible for humans (McCarthy et al., 1980). The AI domain is constantly evolving, as well as our understanding of human intelligence, which can lead to changes in definitions. In the previous century, the founders of AI were unsuccessfully attempting to build computer systems equivalent to human intelligence. To achieve this, they have initiated AI efforts in multiple directions, many of which are still ongoing. In the last century, researchers were applying AI models like heuristic programming, including genetic algorithms and knowledge-based techniques like expert systems, to various fields (Probst & Worlitzer, 1988). All these narrowed applications of AI are called “weak AI”, in contrast to “strong AI”, which is comparable to human intelligence—terms coined by philosopher John R. Searle (1980). A “Weak AI” solves a particular problem, while a “strong AI” would be a generic purpose machine thinking and behaving like a human being.

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An AI-based project schedule creation is considered a “weak AI” solution.

A decade before the term AI was coined, the modern project management era began, and since then, researchers have continuously sought ways to enhance project scheduling methods, aiming to increase the success rate of projects. Over time, this exploration has evolved with the influence of AI, sparking new ideas and approaches within the field of project management. Numerous efforts have been made to apply AI to project management, as evidenced in papers such as (Alshaikhi & Khayyat, 2021; Bodea et al., 2020; Davahli, 2020; Kumar et al., 2021; Mamatha & Suma, 2021; Mishra et al., 2022; Wachnik, 2022; Wijayasekera et al., 2022). The researchers explored how AI can assist project managers and identified potential avenues for further research and development.

The key takeaway is that research progress has not advanced evenly across project management domains. The project management community anticipates that AI will create project schedules autonomously (Fridgeirsson et al., 2021; Gartner Inc., 2018; Holzmann et al., 2022). Currently, the expectations are not only about automatic, but about intelligent autonomous schedule creation based on data gathered in the past and during the current project. Unrealistically high expectations contribute to the failure of project management system adoption (Nichols, 1986).

While AI holds the promise of revolutionizing project scheduling by automating the creation of schedules, there are significant gaps between the expectations and the current capabilities of AI systems. Researchers and project managers alike foresee AI taking over schedule creation autonomously (Fridgeirsson et al., 2021; Mishra et al., 2022; Song & Minku, 2023; Wei & Ehsan Rana, 2019), but the reality has not yet matched these expectations.

At present time, the schedulers, planners, and project managers use their experience and knowledge during the whole project schedule creation process to create a schedule. A schedule is a model for executing the project’s activities enriched with durations, dependencies, and other planning information (Project Management Institute, 2021). Presently, the active participation of practitioners in the schedule creation process is mandatory. They gather schedule data, interpret it, and make informed decisions based on the developed schedule model. Currently available scheduling software accelerates the project schedule creation by schedulers and project managers, but it has not taken the schedule creation burden from them. The next step in the development of IT solutions in project scheduling has become AI-based solutions capable of processing the collected schedule-related data (Holzmann et al., 2022).

Given the critical role of data, the field of artificial intelligence (AI) has grown in significance, particularly with the emergence of machine learning (ML). As AI evolved, ML

became a key subfield, enabling systems to improve and adapt through data-driven learning. This ability to analyze historical project data makes ML a powerful tool for enhancing project scheduling.

ML differs from traditional AI in that it allows systems to improve autonomously by learning from data rather than relying on pre-programmed rules. Arthur Samuel (1959) used a machine learning notion to distinguish deterministic programming from a computer being able to program itself. In other words, ML programs can improve themselves with new experience. Fundamentally, we can say that machine learning is a program’s ability to learn from data. Consequently, ML can use principles of inductive learning, analytical learning, or even combined learning methods (Mitchell, 1997).

Regardless of the learning approach, the principles embraced by ML algorithms serve as guides for acquiring concepts and making inferences. A concept might be a pattern, relationship, formula, or characteristic that the algorithm aims to learn from training data. This input data may or may not have corresponding answers. In other words, the training data set provided may contain independent variables with or without dependent variables.

Machine learning encompasses various methods, with three primary learning approaches: supervised, unsupervised, and reinforcement learning. Supervised learning occurs when an algorithm learns from an example that already has answers. Unsupervised learning is when an algorithm learns from data without providing corresponding answers or suggestions. Reinforcement learning is when a program or agent interacts with the environment to find the optimal way to achieve its goals. Alongside these primary approaches, hybrid approaches are also recognized (Sah, 2020). Machine learning has developed into a distinct field with its own concepts and approaches. For readability and understanding purposes, a summary of essential machine learning concepts is presented in Table 1.

Drawing upon machine learning, organizations gathering project data are interested in the next generation project schedule creation method. Currently, the gathered data are used by planners as pieces of information in project planning processes. The planning knowledge derived from the uncovered pieces of information may not be fully identified due to the limited data processing capabilities of humans. It is challenging for humans to effectively process years of experience accumulated within the data gathered by an enterprise in a reasonably short time. In consequence, business organizations collecting these project data use only a fraction of it in their planning processes (Gartner Inc., 2018). Nonetheless, companies archiving the past project schedule data potentially could train a ML algorithm, where the past projects should be the input for the next-generation project schedule creation. However, it should be noted that there are two important factors for the successful adoption of ML in

Table 1 Essential concepts in machine learning

Concept	Explanation
ML algorithms	A computer procedure that learns from provided data and automatically builds and adjusts a model, which is used for making predictions, classifications, or decisions based on new, unseen data
Training data	Data used to train a machine learning model, consisting of inputs and corresponding outcomes
Testing data	Data used to assess the performance of a trained machine learning model, typically consisting of data that was not part of the training set
Model training	The process of teaching a machine learning model to improve its performance
Supervised learning	A machine learning approach where the model is trained on labeled data to predict outcomes
Unsupervised learning	A technique where the model finds patterns in data without predefined labels or outcomes
Reinforcement learning	An approach where an agent (an ML-based system or algorithm) learns to make decisions by receiving rewards or penalties for actions
Hybrid learning	Combines elements from supervised, unsupervised, and reinforcement learning methods
Feature selection	The process of selecting relevant input variables for use in model building to improve model performance
Intelligent system	A software that applies machine learning techniques to perform tasks autonomously, e.g., creating a project schedule

project scheduling: past data quality and the applicability of specific ML algorithms for a particular problem (Dahmani et al., 2021; Kanakaris et al., 2020; Lee & Shin, 2020; Wen et al., 2012).

The recognition of ML's potential in project scheduling has led to the introduction of the term 'Intelligent System' into project management. According to PMI (2019), a scheduling system is considered to be intelligent when ML algorithms are being applied. However, certain scholars conceptualize the term 'Intelligent System' in a broader context. For instance, they view it as a system that not only uses ML algorithms but also encompasses a spectrum of other planning processes and AI techniques (Shakhsi-Niaei & Iranmanesh, 2015). Furthermore, some researchers use the term 'Intelligent Scheduling,' which refers to the use

of AI techniques to optimize and automate the scheduling of tasks, activities, and resources. This term can be found in publications concerning a variety of industries including manufacturing, transportation, healthcare, logistics, as well as in project management (Fazel Zarandi et al., 2020; Richards, 2012).

In the field of project scheduling with ML, it is evident that a significant degree of confusion and ambiguity spreads through the existing body of literature. This observed lack of coherence within the literature can be attributed to a variety of factors, including divergent industries and a dearth of standardized nomenclature. Furthermore, the view on project scheduling with ML is often fragmented, either confined to a modest corner of the broader discourse on AI in project management or presented in a narrowed manner.

Notably, our investigation found a scarce number of publications dedicated to the use of ML in project scheduling. Moreover, it has been revealed that there is a distinct absence of studies presenting the holistic application of ML in the whole project schedule creation process. There are specialized publications that review only specific areas of schedule creation, like effort estimation (Ali & Gravino, 2019; Sharma & Singh, 2018; Wen et al., 2012) and task assignment (Fatima et al., 2020), while the other areas are neglected.

We address this research gap by conducting a comprehensive review on the possibilities for using ML in all subprocesses of the project schedule creation process. Consequently, the primary aim of this paper is to evaluate the current state of theory and practice regarding the application of ML algorithms in the process of creating project schedules through a systematic review of literature. Specifically, we identify the key areas of project schedule creation where ML is employed and demonstrates the highest potential. Furthermore, our research uncovers the most popular and the most efficient ML techniques when applied to various aspects of schedule creation. We also analyze industries contributing to this research area to gain insight into the advancements made within those sectors. Additionally, we discuss the potential, challenges, and limitations associated with the use of ML in the context of schedule creation. Finally, we explore anticipated advancements, potential benefits, and strategies for effectively integrating ML into project scheduling practices, while outlining implementation paths in the Discussion section.

The rest of the paper is organized as follows: the next section introduces the project schedule creation process, which we use in our review to systematize the literature. The 'Method' section outlines the search strategy and research questions. Next, we present and discuss the results of our study, and finally, we conclude with suggestions for future research.

2 Project schedule creation process

As indicated in the Introduction, one of the objectives of this article is to uncover the specific areas within project schedule creation where ML is the most frequently and effectively applied. Therefore, it is essential to adopt a methodology that divides the schedule creation process into subprocesses. Due to the fact that project schedule management practices are well established and documented, we have decided to use one of the commonly accepted standards for the classification of project scheduling approaches provided and documented by some of the most known and internationally recognized organizations, such as the International Project Management Association (IPMA), the Association for the Advancement of Cost Engineering International (AACE), the Association for Project Management (APM), the International Standardization Organization (ISO), and the Project Management Institute (PMI). The government-type publications, including the schedule management aspects, were excluded from the analysis due to their limited applicability.

Ultimately, our choice leaned toward the approach documented by the Project Management Institute (PMI) in their “Practice Standard for Scheduling” (PMI, 2019). This decision is based on the document’s detailed and comprehensive nature, clarity, consistency, and its well-established recognition within the project scheduling community. It is important to note that other standards offer valuable insights and methodologies. However, for our research goals, we found that the PMI standard facilitates the detailed and generic subdivision of the project schedule creation process. We believe that our study benefits from the detailed description of schedule creation steps provided in the ‘Practice Standard for Scheduling,’ which aligns well with our classification purposes. Moreover, PMI also documents the scheduling process in other publications like “Project Management Body of Knowledge” (PMI, 2021). However, “Practice Standard for Scheduling” provides a higher level of granularity and has a more detailed description of schedule creation steps, which supports classification purposes better. Furthermore, the chosen standard does not focus on any specific scheduling method nor industry-specific aspects.

Furthermore, academic research on project scheduling is divided into two main areas: resource-constrained scheduling and scheduling with the assumption of resource availability (Vanhoucke, 2013). The generic procedure proposed by PMI is agnostic to this classification, as it does not impose a specific sequence of subprocesses. Moreover, the process framework recognized by PMI does not dictate whether it is used in predictive or adaptive planning. The sequence of schedule management processes according to PMI can vary, and the extent of their implementation depends on the project methodology and the specific tailoring of project processes. Our research focuses on project schedule creation in general,

with the aim of classifying ML usage from the perspective of different scheduling steps. We consider that the ML technique used in one type of schedule techniques can be used in others. However, when presenting results, we recognize the type of scheduling being used, whether it is resource-constrained or not.

Though interpretations may vary, here the schedule creation process includes both the schedule model and the project schedule itself. A schedule model is a representation of the plan for executing the project’s activities, including duration, dependencies, and other planning information (PMI, 2019). The additional information includes resource requirements, schedule constraints, and any other details that help the project team predict project completion dates through schedule model instances. Thus, a project schedule is an output of a schedule model that presents linked activities with planned dates, durations, milestones, and resources (PMI, 2021). In accordance with PMI scheduling standards, this study uses the following categories of the schedule creation:

- milestone definition,
- project activities definition,
- project activity sequencing,
- project activity resource determination,
- project activity duration determination,
- schedule optimization.

PMI schedule model creation also encompasses: schedule approval, baselining, and schedule levels creation, each of which plays a distinct role in the process. However, these process steps are not currently under consideration for integration with ML techniques. The schedule approval is a solely human activity; therefore, it is not possible to replace a human with AI at its current development stage. Establishing a project schedule baseline provides a snapshot based on the approved schedule model, where ML is unlikely to add value. Schedule levels are visualizations of the created schedule model used in stakeholders’ communication. Therefore, the schedule-level creation is a view of the currently established schedule model, where the ML algorithm is not required.

Milestones are important for control and communication mechanisms. They represent significant events or points in the project (PMI, 2021). It can also be said that a milestone is a specific type of activity which has zero duration, and no resources assigned (PMI, 2019). Milestones are important for sharing progress with broader and senior stakeholders. Each project has start and finish milestones, but the rest of the milestones depend on organizational preferences, external constraints, and internal control points, like approvals or quality gates (APM, 2015).

Project activities definition provides the list of activities. Traditionally, the list is created by the project team

that decomposes the Work Breakdown Structure (WBS) into activities (PMI, 2019). During the schedule creation process, the knowledge about activities composing the schedule is growing. The data characterizing and defining activities are called activity attributes (PMI, 2019). Creating a list of activities is one of the key aspects of the project activities definition. Once the activity is identified, its characteristics are discovered. From the schedule model perspective, the important activity characteristics are: relationships with other activities and milestones, resources required, constraints, and duration. Traditionally, the domain-specific activity attributes are elicited through expert knowledge aggregated within the project team. All identified activity attributes are used in the next steps of the schedule creation process.

Project activity sequencing output is a logic explaining the relationship between activities and milestones. The logic explains activity succession and precedence, relationship types, constraints, leads, and lags (PMI, 2019). One of the popular ways to present this logic is a graph, called a network diagram (PMI, 2021), which represents activities, milestones, and relations between them. One of the popular graphical techniques of project activity sequencing is Activity on Node (AoN) (APM, 2015), in which graph nodes represent activities, and relationships are visualized by links. Therefore, the activity sequencing addresses graph-related problems, such as identifying links, determining types of relationships between nodes, and identifying other link characteristics.

During **activity resource determination**, for each activity, the type and quantity of labor, materials, equipment, or infrastructure required is identified (PMI, 2019). For example, labor units are expressed as activity effort, which is the number of units required to complete a scheduled activity (PMI, 2019). The effort level required can be expressed in man-hours, man-days, or FTE (full-time equivalent). In fact, the next subprocess requires effort level value to properly estimate activities duration (PMI, 2021). Activity effort is a result of a quantitative assessment of its likely amount, which is called effort estimate (PMI, 2021). There are various techniques to estimate activity effort that are listed in PMI, APM, and IPMA publications (e.g., Bradbury et al., 2019; PMI, 2021). From a practical point of view, we can organize known effort estimation techniques into the following categories: expert judgment, bottom-up approach, analogues estimation, parametric estimation, and data analysis (PMI, 2021). All the broadly known effort estimation techniques rely on the processing ability of a human being—even if supported by the current computer programs. Before estimating labor effort, the identification of resource types is necessary. An activity can require various types of resources. Moreover, the classification of resources is not a trivial problem. First, resource types can be industry-specific. Furthermore, enterprises have their own resource type definitions. Finally,

the inherent nature of the activity and its chosen approach determine the types of resources required.

Determination of project activity duration establishes the total number of working periods required to complete a schedule activity (PMI, 2021). This process also should consider specific activity attribute information discovered in the previous steps. Duration estimation is a category covering qualitative assessment techniques used to provide the likely number of calendar days, weeks, or other time periods required to execute an activity. An activity, task, or project has a duration which is determined by its start and its end. The units used are calendar days, weeks, months, or even years. Duration estimates can also be a range of possible intervals (PMI, 2019). It should be noted that the duration of an activity depends on various factors, like holidays, working culture, time allocation, and many others, including an effort required which can seriously impact duration. A human resource's working time can be full-time or part-time, either for the entire project duration or a limited period, and the work can be collocated or virtual, depending on the environmental factors (PMI, 2021). In fact, an activity may have a range from a few-hour job to a project-size work. There is no standardized definition of what an activity is and what the size of an activity is in project management. An activity can be a few-hour task performed by one person or a complex piece of work carried out by a group of people using human, material, and other resources.

Schedule analysis and **optimization** is the last step performed before the target schedule model is created. When this step of project schedule creation is achieved, it is clear what work needs to be accomplished and what activities are expected to be performed to achieve the project (PMI, 2019). During this process step, the schedule model is updated with dates. Notably, the schedule is optimized to ensure that project objectives and constraints are met. Traditionally, during this step, resource leveling or smoothing is being done by a project scheduler (PMI, 2021). Commonly, if project time constraints are not met, then crashing or fast tracking is applied to the project schedule (PMI, 2019).

Additionally, an important aspect in schedule model creation is its improvement through the application of risk management. The schedule model improvement is an important aspect of project schedule resiliency (Dahmani et al., 2021); however, this area is beyond our study.

3 Method

The project schedule creation process is a wide area covering many activities explained in the previous section. This increases the complexity and breadth in the review process. Moreover, the junction of machine learning and project management covers emerging and evolving notions. This implies

a complex and comprehensive review involving the analysis of literature, which encompasses domains such as project management, artificial intelligence, data science, and data mining.

Systematic review allows transparent presentation of emerging knowledge with minimization of personal bias and identification of gaps in the current knowledge (Tranfield et al., 2003). Due to that fact, we have chosen the systematic literature review rigor suggested in (Denyer & Tranfield, 2009), which addresses specifics of management fields. Tranfield's approach allows a pragmatic approach that helps with the complexity of the research objective. The next subsections document all decisions made for this systematic review.

3.1 Research questions

This study is guided by five research questions that explore how machine learning (ML) is applied to project schedule creation and what challenges and opportunities arise from this integration.

We begin by asking **RQ1: “In which project scheduling subprocesses is ML used?”** This question lays the groundwork for evaluating research maturity, identifying remaining gaps, and understanding the specific areas where ML contributes to improving the schedule creation process.

Based on this, **RQ2: “Which ML techniques are used in project schedule creation subprocesses?”** investigates which specific algorithms and models are employed—particularly those most frequently used—in different areas of the schedule creation. Moreover, this question offers an opportunity to explore the qualities that make certain ML techniques particularly valuable for distinct schedule creation subprocesses.

Next, **RQ3: “Which industries have the highest number of research studies on ML in project scheduling?”** shifts the focus toward the application domain. Identifying the most represented industries allows us to infer where ML-driven scheduling has gained the most traction and can help indicate which sectors are most likely to adopt ML in project scheduling.

Effectiveness is at the core of **RQ4: “What is the effectiveness of ML techniques in the project scheduling process?”** This question seeks to provide aggregated evidence on how well ML techniques perform in this context. It also helps uncover the potential advantages of applying ML to project scheduling and assess the applicability of specific techniques within the schedule creation process.

Finally, we broaden the scope with **RQ5: “What are the benefits, challenges and limitations, and recommended strategies for the effective application of ML techniques in project scheduling?”** This question aims to examine the effective use of ML in project schedule creation from a broader perspective. It focuses on identifying recent

advancements that could significantly influence the scheduling process, understanding the obstacles that hinder broader adoption, and uncovering strategies for successful integration. Ultimately, RQ5 seeks to synthesize current knowledge to support more informed and practical applications of ML in scheduling.

Together, these questions form a cohesive framework for analyzing the current landscape of ML in project schedule creation and guiding future directions for research and practice.

3.2 Research strategy

To ensure the replicability of this review, a search strategy is provided. The first attempts to use ML in project schedule management started in the 1990s (Wen et al., 2012 and the reference therein); however, some AI solutions, including metaheuristics, expert systems, and evolutionary algorithms, date far back. Therefore, the search queries were limited to the literature from the year 1990 to 2023. This period reveals the trends and ensures comprehensive coverage of ML since the beginning of research on this topic.

The databases used in this review are Scopus and Web of Science. The Scopus database covers high-quality project management, software engineering, and artificial intelligence journals. Web of Science is another widely accepted and well-known publication database; therefore, it was used to cross-check the search results and increase the quality of our review. The search is performed on ‘title’, ‘abstract’, ‘author keywords’, and ‘indexed keywords’ generated by the platforms.

This research is focused on ML in project scheduling, but other keywords used by researchers may cover important content related to this topic. This includes terms like ‘artificial intelligence’, ‘intelligent systems’, ‘intelligent scheduling’, ‘data science’, ‘data mining’ and the names of specific algorithms. Furthermore, ‘keywords’ can be combined with various additions like ‘application,’ ‘techniques,’ and many others. Thus, the search conditions below include related terms with wildcard characters to ensure the search engine also accepts more sophisticated keywords.

We applied two queries to identify the literature sources. The first query aims to identify the broad range of papers where the project scheduling is a part of the general discussion related to the artificial intelligence techniques’ expansion into the project management domain. The second query aims to identify papers strictly related to project scheduling with ML techniques. It uses terms that are typically used in project scheduling.

First Query

("*project management*" OR "*project plan*" OR "*project sched*" OR "*project time*") AND ("artificial

intelligence*" OR "*machine learning*" OR "*machine-learning*" OR "*ML algorithm*" OR "deep learning" OR "neural network*" OR "data science" OR "intelligent system*" OR "predictive analytics" OR "data mining" OR "learning algorithm*" OR "ensemble learning" OR "clustering algorithm*" OR "tree algorithm*" OR "decision tree*" OR "nearest neighbor*" OR "random forest*" OR "reinforcement learning*" OR "unsupervised learning*" OR "supervised learning*" OR "intelligent schedul*" OR "feature selection" OR "support vector*" OR "case based reasoning*" OR "case-based reasoning*" OR "gradient boosting*").

Second Query

("activity duration*" OR "task duration*" OR "activity effort" OR "task effort" OR "decomposition" OR "activity resource*" OR "task resource*" OR "define activities" OR "define tasks" OR "duration of activities" OR "duration of tasks" OR "fast tracking" OR "milestone*" OR "precedence relationship" OR "predecessor*" OR "resource type" OR "resource leveling" OR "resource assignment" OR "schedule model" OR "sequence activit*" OR "successor*" OR "float" OR "slack") AND "project*" AND "schedule*" AND ("*machine learning*" OR "*machine-learning*" OR "*ML algorithm*" OR "deep learning" OR "data science" OR "intelligent system*" OR "data mining" OR "learning algorithm*" OR "ensemble learning" OR "reinforcement learning*" OR "unsupervised learning*" OR "supervised learning*" OR "intelligent schedul*").

3.3 Selection and evaluation

A set of explicit criteria was defined to assess the relevance of each identified paper. These cover the inclusion and exclusion criteria documented in this subsection. The steps applied in the process of paper selection and evaluation, as well as their results, are shown in Fig. 1. First, we executed the designed queries on the database and exported results (step 1). Secondly, we merged the paper lists and deleted duplicated records to understand the total number of unique papers (step 2). Next, we applied the exclusion criteria on the extracted papers to speed up the selection and the evaluation process (step 3). Finally, we selected papers by applying the full inclusion and exclusion criteria to identify all eligible studies, including those providing partial answers to our research questions (step 4).

In addition, Fig. 1 shows a split of the papers into two categories: 'primary research' and 'other' papers. 'Primary research' refers to scientific papers that present original research findings, data, or analysis. These papers are based on direct investigation, experimentation, or observation conducted by the authors, for example: reports on original experiments, case study papers, comparative studies. The

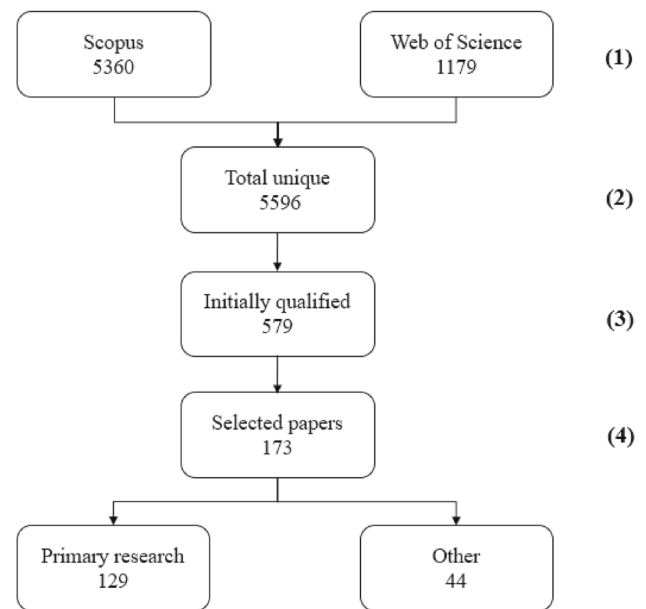


Fig. 1 Selection and evaluation steps with corresponding number of papers

'other' papers cover various types of reviews and publications that partially address research questions which are not directly focused on the field studied. The list of primary research papers is accessible online at <https://tinyurl.com/MLinProjectScheduleCreation> for any further investigation. In our review, we cited and referenced the primary sources directly used.

3.3.1 Inclusion criteria

We applied the following inclusion criteria to identify studies that specifically address ML aspects in project scheduling:

- studies that use ML in project schedule planning activities,
- studies that compare different ML techniques with regards to project scheduling,
- studies that discuss the potential of using ML in project schedule creation,
- studies that theorize on the application of ML in project schedule creation.

3.3.2 Exclusion criteria

We applied the exclusion criteria to identify which studies should be removed from further analysis. The criteria cover:

- studies that elaborate solely on scheduling in non-project domains like: job scheduling systems, production, manufacturing,

Table 2 Number of research papers according to project schedule creation subprocess

Subprocess	No. of studies
Milestone definition	0
Activities definition	1
Activity sequencing	5
Activity resource determination	101
<i>Effort estimation</i>	95
<i>Resource allocation</i>	6
Activity duration determination	15
<i>Activity duration</i>	2
<i>Project duration</i>	13
Schedule optimization	6

- studies that elaborate on project scheduling but do not mention or address ML aspect,
- non-English papers.

If a study was published both in journals and conference proceedings, only the journal version was considered.

4 Results

In this section, we present the findings of our review. First, we highlight and analyze selected studies, organizing them according to the project scheduling subprocesses explained in Sect. 2. Next, we address the research questions based on our findings.

4.1 Overview of key findings per subprocess

Table 2 summarizes the number of studies on each project creation subprocess. In addition, the table shows areas of scientific research associated with the most widely studied subprocesses. It is clearly visible that the most explored area is effort estimation, which is part of the activity resource determination subprocess. The findings related to each subprocess are detailed in the corresponding subsection.

4.1.1 Milestone definition

We have not identified any study supporting ML in milestone definition. The problem of identifying milestones seems to be similar to the problem of identification of activities. Additionally, a milestone may not only come from past project data, but also from other organizational assets which can impose milestones in a prescribed way. Furthermore, project milestones could be deducted from a statement of work, contract documentation (Holzmann et al., 2022). Despite numerous

potential possibilities, no results were identified on this subject.

4.1.2 Activities definition

Machine learning algorithms used for project activities definition should provide more than just the activity's name as an output. It should also provide the activity attributes, which are important inputs for all the next steps of the schedule creation process. In fact, we found only one study that addresses the need of creating a list of activities with regard to the resource-constrained project scheduling problem (Gómez & Plaza, 2012). The study proposes the usage of the Case Base Reasoning (CBR) algorithm to retrieve lists of activities from similar historical projects. The CBR-based solution seeks a similar project based on characteristics provided by the user, using the classical k-nearest-neighbor algorithm (k-NN). The sequencing of the activities is handled via a resource-constrained scheduling method called Serial Schedule Generation Scheme, which is not in the focus of our review. Gómez and Plaza (2012) noticed that the CBR-based solution provides better results than the heuristic-based solution that they used for benchmarking. Furthermore, the authors suggest that this approach serves as a valuable option for managing the trade-off between solution quality and efficiency. As stated by the authors, the discussed CBR solution does not guarantee the best result for specific scenarios; however, it provides a competitive solution in a wider range of problems, such as those characterized by poorly defined domains or high degrees of uncertainty. Moreover, this solution addresses a specific scheduling problem where resources are constrained. Furthermore, the CBR-based solution improves a given project schedule, and it does not define its activities from scratch. This solution provides indications of how an activities list can be identified from historical data gathered by an organization. It is worth adding that the CBR algorithm has learning capabilities that can support new knowledge discovery (Mitchell, 1997). Although CBR learning was applied by Gómez and Plaza (2012) to randomized existing cases, the authors claim that the continuously enhanced CBR-based solution has the potential for good efficiency in tackling complex and uncertain real-life scheduling problems.

The relevant attributes of identified activities should also be discovered. The quality of all subsequent steps in the project schedule creation relies on the availability and accuracy of the activity attributes provided. These are important input parameters for an expert judgment or computer-aided software, including ML. From a methodological point of view, the activity attributes can be regarded in supervised learning as 'features', and as such, they can be used for prediction purposes in the next schedule creation subprocesses. It is well-known knowledge that features have a significant

impact on the investigated algorithms and studies' results (Mamatha & Suma, 2021). The feature discovery and selection are complex processes that require thorough analysis when planning experiments and considering the adoption of ML in the schedule creation process. However, we did not find any study strictly focused on the identification of activity attributes.

4.1.3 Activity sequencing

In the project activity sequencing subprocess, depending on the type of scheduling problem, a ML-based sequencing process can link activities using the most natural 'finish-to-start' (FS) relationship. Alternatively, it could create a complete schedule network with other types of relationships: 'start-to-start' (SS), 'finish-to-finish' (FF), and 'start-to-finish' (SF). ML-based solutions could also determine if a particular relationship between activities is soft or hard logic, which is an important factor during schedule optimization.

We have found five papers on ML adoption for this subprocess. Four identified studies propose the usage of artificial neural networks (ANN) to determine activity sequencing (Alikhani & Jeong, 2022; Amer & Golparvar-Fard, 2019, 2021; Golpayegani & Parvaresh, 2011). Amer and Alikhani present results from their investigation of long short-term memory recurrent neural networks (LSTM-RNN) to resolve activity dependency and sequence logic from past projects. Amer and Golparvar-Fard demonstrated advancements in the use of LSTM-RNN within their developed Dynamic Means and Methods Templates (DMMTs) method. The authors defined DMMTs as a method that is able to formalize sequencing logic and mine from previous projects to provide a basis for schedule manipulation and revision. The authors suggest various applications of the DMMTs method, such as logic validation and schedule generation, which could serve as extensions for a commercial product. Golpayegani and Parvaresh (2011) studied Multilayer Perceptron Neural Network (MLP-NN) for FS sequence identification. The experiment documented in (Golpayegani & Parvaresh, 2011) relied on the MLP-NN algorithm trained from data related to a specific project domain. This study assumed only FS relationships, which limits the idea presented to only FS-type schedule problems. The real-life project requires more complex relationships. However, Golpayegani and Parvaresh (2011) noted that when identifying multiple activity relationships, it becomes necessary to scale up the constructed neural network, which in turn requires greater computing power. Still, the authors do not provide any results for a more comprehensive activity sequencing process. Ryoo and Ashtab (2020) propose a solution based on the Frequent Pattern Growth (FP-growth) algorithm, which discovers association rules explaining relationship types (FF, SS, SF, FS) along

with their corresponding lags. The authors prove that comprehensive activity sequencing logic can be learned by ML and applied later in the schedule creation process. The FP-Growth algorithm discovers association rules between activities and identifies recurrent patterns that can then be applied to a new project. In this way, various types of relationships and related lags can be discovered. The authors indicate that their solution has potential in generating alternative schedules, which can support schedule optimization.

In fact, all identified studies address objectives of activity sequencing subprocess, and they can be used for various scheduling problems. However, only FP-growth provides a method that is sufficiently comprehensive, and what is also important, was verified for FS, SS, SF, and FS relationships with lags. Amer and Golparvar-Fard () and Golpayegani and Parvaresh (2011) are aware of the limitations of their experiments on NN and suggest further studies. Moreover, DMMTs method of Amer and Golparvar-Fard is designed for a specific usage scenario where LSTM-RNN plays a role in sequencing. However, DMMTs method applicability is limited. The aforementioned studies highlight that a schedule can be viewed as a graph, which is created by establishing relationships among activities. The authors of the referenced studies noticed that the sequencing problem is an expert knowledge domain problem, but ML techniques can provide more flexible solutions.

4.1.4 Activity resource determination

Activity resource determination can involve several distinct ML solutions, addressing various aspects of this subprocess, including determining the type of resources and required resource quantity.

We did not identify any technological categories related to material, equipment, or infrastructure estimation. Therefore, this section presents results on using ML for determining human resources. The human resources determination consists of two aspects: identification of specialists or roles required to execute a particular activity, and effort estimation.

We found that the first aspect is addressed in the studies dedicated to the resource allocation. Fatima et al. identified the applicability of support vector machine (SVM), naive Bayes (NB), decision tree (DT), regression tree (RT), k-nearest neighbors (k-NN), and logistic regression models to resource allocation (Fatima et al., 2020). We also found studies on assigning the most suitable person for the execution of a specific activity (Anvik et al., 2006; Kanakaris et al., 2019, 2020; Samath et al., 2017). All four mentioned studies use natural language processing (NLP) with ML. Moreover, Shahhosseini and Sebt (2011) propose an adaptive neuro-fuzzy inference system (ANFIS) that takes into account the required competences to perform the activity.

Our study provides evidence supporting that the effort estimation is the most popular researched category among other project scheduling areas (see Table 2), which has also been noticed by other researchers (Ali & Gravino, 2019; Chennouk et al., 2020; Sharma & Singh, 2018). We identified three systematic studies on effort estimation (Ali & Gravino, 2019; Sharma & Singh, 2018; Wen et al., 2012), four systematic mapping studies (Amazal & Idri, 2019; Boujida et al., 2021; Hosni & Idri, 2018; Idri et al., 2016), and also three narrative review papers (Dhanda & Sangwan, 2017; Kumar & Srinivas, 2021; Mendes, 2010). Furthermore, the effort estimation was detailed in other more generic papers on ML in project management (Chennouk et al., 2020; Davahli, 2020; Mahdi et al., 2021). This category is significantly explored, especially in the software engineering domain, where the effort estimation is considered one of the key aspects of project management.

The results of our study confirm that ANN is the most popular algorithm in effort estimation, since seventy papers on this topic had been identified. Some researchers concluded that ANN outperforms other ML algorithms used for effort estimation purposes (Ali & Gravino, 2019; Chennouk et al., 2020; Sharma & Singh, 2018; Wen et al., 2012). The other popular ML methods in this area are SVM (Ali & Gravino, 2019) and fuzzy logic-based ML algorithms (Sharma & Singh, 2018) and CBR (Wen et al., 2012). However, it might be impossible to univocally conclude which algorithm is the best for effort estimation, since analyzed studies use different datasets and different performance metrics (Mendes, 2010). There is general agreement that different ML techniques vary in terms of strengths and weaknesses depending on various circumstances (Wen et al., 2012). Wen et al. noticed that ML algorithms' efficiency depends on dataset characteristics, like its size, noisiness, outliers, missing values, and the chosen features. For example, CBR and BN algorithms are less sensitive to small data sets (Wen et al., 2012) (Wen et al., 2012), which may be significant if an organization has a small amount of past project data. In turn, ANN is capable of dealing with noisy data (Wen et al., 2012). This might be important if dealing with a large number of data, where proper data cleansing is too expensive or time-consuming. Moreover, it is claimed in the literature that the most preferable approach to achieve accurate effort estimation is to apply ensemble learning, which consists of using a combination of ML algorithms (Dhanda & Sangwan, 2017). Experiments conducted in (Hosni et al., 2017, 2018, 2021; Jodpimai et al., 2018; Pai et al., 2013; Sakhravi et al., 2021) prove that ML ensemble methods provide improved effort estimation results.

Lastly, it should be mentioned that the overwhelming number of effort estimation papers comes from the software engineering domain. These researchers concentrated

on applying ML to inputs commonly employed in well-established parametric estimation methods, such as the COCOMO model. Generally, these studies utilized datasets specifically designed for those methods. Thus, the machine learning experiments used the same variables required by the parametric estimation method, and no further work was done on the selection of variables to improve the effectiveness of ML algorithms. However, the significance of variable selection has been recognized in the field of effort estimation, and this is recognized in a systematic mapping study by Idri (2016).

4.1.5 Activity duration determination

We have identified two papers strongly related to activity duration determination using ML (Gvozdenović, 2007; White & Awadallah, 2019), even though the authors of these papers did not explicitly position their work in the area of project activity duration determination. Furthermore, there are a considerable number of papers available on project duration estimation using ML techniques that could be leveraged for activity duration determination. However, none of the identified studies address the activity duration determination process literally.

White and Awadallah (2019) analyzed personal task duration estimation using ML algorithms trained on past calendar data. Logistic regression and neural network algorithms were trained on a sample of millions of calendar events. This study clearly documented the example of feature selection, which included content, context, and history attributes. It reports a positive impact on the accuracy of duration estimates when features are extracted from the content. Furthermore, the study concluded that deep learning techniques are more accurate than logistic regression.

Gvozdenović (2007) sought a Program Evaluation and Review Technique (PERT) extension with ANN applied to estimate activity duration from optimistic and realistic scenarios. The study tested the feed-forward backpropagation network (FFBP) for activity duration estimation in the PERT method. The obtained results confirm the usefulness of NN, which is notable because PERT is a widely accepted method that can provide ensemble guidelines in activity duration determination.

The studies dedicated to estimating the whole project duration can also provide relevant insights into the application of ML to activity duration. In fact, the features describing activity can closely resemble those of an entire project. We have identified 13 studies which, in general, checked the feasibility of ML algorithms for project duration estimation (Burhan et al., 2021; Chen & Huang, 2006; Cheng & Hoang, 2017; Guo et al., 2019; Hamada et al., 2021; W. Han et al., 2014; Han et al., 2015a, 2015b; Laqrichi et al., 2014; Li et al., 2017; Liu et al., 2022; Pospieszny et al., 2018; Relich

& Muszyński, 2014; Yau & Yang, 1998). Nearly all identified papers used various types of NN; some of them also inspected SVM, DT, KNN, CBR, and linear regression algorithms. It can be concluded that ML techniques discussed in the referenced studies offer viable alternatives to the traditional project duration estimation methods.

4.1.6 Schedule optimization

Machine learning finds applications also in optimization, where reinforcement learning techniques are particularly prominent (Mitchell, 1997). This suggests that reinforcement learning is also likely to establish itself in schedule optimization. The schedule model constructed in proceeding steps can undergo analysis using an ML technique to identify potential schedule instances that meet the objectives defined by the project planner. ML potentially could optimize a schedule to meet resource constraints, deadlines, cost objectives, or even take into account risk factors. However, due to the scope of this review, our search queries specifically do not address the aspects of schedule risk management with ML. As a result, we do not provide details or analysis regarding the risk management aspects undertaken as part of schedule analysis. This is out of scope for this review.

In the 1980s, there were attempts to apply neural networks to resource leveling (Alsugair & Chang, 1994). We revealed that in the period considered in our study, i.e., from 1990 to 2023, there were no further attempts to address resource leveling with ML. This is consistent with observations that nowadays, suboptimal resource leveling is still accomplished using the current software controlled by a planner (Richards, 2012).

In the area of resource-constrained project scheduling, we found studies that apply reinforcement machine learning to optimize schedules. Three studies (Jedrzejowicz & Ratajczak-Ropel, 2014; Mirzaei & Akbarzadeh-T., 2012; Wauters et al., 2011) used the Multi-Agent Reinforcement Learning (MARL) technique to minimize total project duration successfully. Sung et al. (2020) used a solution based on deep reinforcement learning (deep RL) where resource allocation was optimized. A novel approach was presented by Kedir et al. (2022), who proposed a solution based on reinforcement learning with a graph neural network (GNN) supporting a rewarding mechanism. Kedir et al. proposed an advanced ML approach and shared the results of a few simple case studies. The authors conclude that reinforcement learning and agent-based modeling (RL-ABM) with graph embedding presented is a potential solution for schedule optimization. In fact, resource-constrained scheduling does not address all possible project scheduling problems, but the application of reinforcement learning techniques to schedule optimization looks promising and can be applied to other scheduling problems.

4.2 Answers to the research questions

Below, we provide answers to each research question formulated in Sect. 3.1. The synthesis and implications of the findings are discussed in the Discussion section, along with proposals of options for implementation, the highlights of research gaps in the field, and the suggestions for further study.

4.2.1 RQ1: In which project scheduling subprocesses is ML used?

We found ML papers concerning all project schedule creation subprocesses except for the process of milestone definition. However, it is evident that the project creation subprocesses have not received equal attention. We did not identify any studies regarding milestone identification, even though ML has the potential to assist by processing past project milestone data and identifying relevant milestones for the project context.

Machine learning can be an aid in creating activity lists within the project activity definition subprocess. However, a comprehensive application of ML to identify activity attributes was not found in our study. We identified a small number of papers addressing activity feature selection, which can provide insights for attribute identification; however, more studies are needed in this area.

Although we found a few proposals to address the needs of the project activity sequencing subprocess, only one study offered a comprehensive solution to identify all types of relationships. We identified no papers covering ‘leads and lags’ or addressing the concepts of ‘hard and soft logic’.

The activity resource determination subprocess has a significant number of studies focused essentially on the application of ML in effort estimation. ML-based effort estimation has received considerable attention since the early 1990s. The existing systematic reviews in the field of effort estimation, along with our own review, confirm the significant attention and research that has been devoted to this area. However, we did not find any direct studies specifically addressing the identification of resource types. Nonetheless, there is potential for research in this area, which could focus on organizational assets or competency frameworks specific to certain industries.

Activity duration determination is an area that has received limited attention from the researchers. In our study, we found only two studies that have an indirect connection to this subprocess. However, we think that related studies in effort estimation and total project time estimation can serve as a solid basis for further exploration of activity duration determination. All studies indicate that reinforcement learning is the ML field that holds the greatest potential for benefiting the schedule optimization subprocess. In our research, we

encountered a study within this area that proposed the usage of graph neural networks, still within the reinforcement learning context.

4.2.2 RQ2: Which ML techniques are used in the subprocesses of project schedule creation?

In our study, we have found that different ML algorithms are applicable in individual subprocesses. When it comes to activity definition, only the CBR technique has been explored as a relevant ML technique. In the activity sequencing subprocess, there are available papers on ANN and one on FP-Growth, which provide the most comprehensive solutions. In the area of effort estimation, the most popular applications are neural-network-based, followed by decision tree-based and support vector machine applications. Regarding the activity duration determination, artificial neural networks again dominate the field. Figure 2 summarizes ML algorithms supported by research papers.

4.2.3 RQ3: Which industries have the highest number of research studies on ML in project scheduling?

The overwhelming majority of studies come from software engineering, mostly due to the importance of effort estimation in software development and its huge impact on determination of software development cost. We have also identified a dozen studies from the construction industry. In our research, we identified individual papers that focused on the energy and military industry.

4.2.4 RQ4: What is the effectiveness of ML techniques in the project scheduling process?

The majority of studies promote analyzed ML techniques. Generally, they showed that a well-trained ML algorithm can be suitable for the specific schedule creation subprocesses and can outperform the alternative methods. In our research, we discovered only two studies that explicitly expressed a negative opinion about ML, suggesting that human-based techniques (Ionescu, 2017) or statistical methods are superior for effort estimation (Han et al., 2015a, 2015b). The vast majority of researchers have confirmed the positive impact of ML on the accuracy of effort estimation using various error validation methods. In general, the majority of researchers provided positive feedback on the application of ML techniques in project schedule creation.

Based on the analyzed literature, we can draw the following specific conclusions regarding the effectiveness of ML algorithms:

- A. CBR applied in project activity definition can provide reusable activity lists.
- B. FP-Growth algorithm is capable of uncovering rules to establish activity relationships and create a network diagram as part of the activity sequencing subprocess. ANNs require scaled architecture to provide similar results.
- C. Machine learning prediction techniques including ANN, DT, SVM can be effectively used in resource determination and activity duration determination.
- D. Reinforcement learning like MARL, RL-ABM, deep RL can effectively optimize schedules, particularly when considering resource constraints scheduling problems.

Undoubtedly, the capability of ML to process large amounts of data surpasses the effectiveness of a human brain. However, despite the limited number of studies, in most areas, it is difficult to draw categorical conclusions about the superiority of particular methods in specific schedule creation subprocesses. Moreover, the studies analyzed confirm that ensemble learning is a promising ML approach, which should be developed and further investigated in the context of project scheduling.

4.2.5 RQ5: What are the benefits, challenges and limitations, and recommended strategies for the effective application of ML techniques in project scheduling?

The consensus among researchers is that ML can significantly improve scheduling practices, including aspects such as accuracy and efficiency. However, ML application remains underexplored across many subprocesses of project scheduling. While the effort estimation area has been the subject of relatively numerous studies, other aspects of scheduling receive limited attention. Consequently, significant uncertainties persist regarding the broader potential of ML in schedule creation.

The current challenge is not only an uneven focus on scheduling areas but also a lack of effort in exploring ML's overall application in project scheduling. Although more comprehensive studies exist, such as those on DMMT and CBR-based solutions, they only propose the usage of ML in individual aspects. We have not found a study that offers an integrated strategy for broader ML adoption in the schedule creation process. This absence of comprehensive frameworks and actionable strategies poses a significant practical challenge for organizations seeking to effectively integrate ML into project scheduling. Our thoughts on potential paths forward are discussed in the next section.

A common challenge faced by researchers is the limited number of real-life datasets available for scheduling studies. Datasets that are available to the public often contain a narrow set of variables that are tailored to specific, well-known methods like COCOMO. This restricts the scope of potential

Fig. 2 Overview of confirmed ML techniques for project schedule creation subprocesses.

Note: ML algorithm abbreviations are introduced alongside their full names upon first mention in the text

Activity definition	Activity sequencing	Activity resource determination	Activity duration determination	Schedule optimization
▪ CBR	▪ ANN (LSTM-RNN, MPL-NN) ▪ FP-Growth	▪ ANN ▪ DT ▪ SVM ▪ NB ▪ RT ▪ k-NN ▪ ANFIS ▪ CBR	▪ ANN (e.g. FFBP, DNN) ▪ SVM ▪ DT ▪ KNN ▪ CBR	▪ MARL ▪ RL-ABM ▪ Deep RL

experiments and may prevent the discovery of more effective ML applications.

5 Discussion

Current academic research is not keeping pace with community expectations for full automation of schedule creation. Although research has made progress, no comprehensive ML-based method yet addresses all steps of the schedule creation process. Research efforts have focused on applying selected ML techniques to aspects of specific subprocesses in schedule creation, such as activity definition, activity sequencing, resource determination, duration estimation, and schedule optimization. Significant gaps persist, highlighting the need for further studies across all subprocesses, which are explored and discussed in greater detail in this section, along with practical recommendations for implementation.

Specifically, we identified only one study focused on activity definition, where the CBR algorithm was used to support project activity generation. A related area, milestone definition, remains entirely unexamined. In our opinion, artificial neural networks could significantly support both subprocesses. For example, a neural network algorithm could learn to provide a list of activities required by a project or generate a schedule template that can be improved by a project scheduler. The learning process could rely on historical project activities, but also on project context information.

Another approach for project activity generation could involve natural language processing (NLP). In fact, the majority of project data is textual, and we have not identified any studies that leverage NLP in the context of project schedule creation. Given the increasing capabilities of NLP, this area represents a promising direction, where understanding project context and documents is critical. An NLP could process project documentation to generate a list of project activities that can be further updated by a project team.

Generally, NLP could also support other schedule creation subprocesses. NLP preprocessing could support feature

extraction, which could then be combined with selected ML techniques. For instance, by analyzing a project's context description, an NLP-based algorithm could classify the project and identify key characteristics that may serve as input variables for the ML algorithms used in an ML-based scheduling method.

Discussing activity sequencing, despite the limited number of studies, current research provides comprehensive implementation options, such as various ANNs and the FP-Growth algorithm. However, researchers have yet to explore other ML algorithms, such as CBR, decision trees, or other rule-based approaches, which, in our opinion, offer the potential to support the objectives of the activity sequencing process. For instance, CBR could analyze the relationships between activities in historical project data to suggest relevant dependencies for new projects. Similarly, decision tree algorithms could assist in the decision-making process for activity sequencing by providing rule-based pathways that link activity dependencies. These are promising directions for future studies that are notably absent in the current literature. From a practical perspective, leveraging these approaches could help planners establish sequences aligned with proven workflows or eventually automate the assignment of relationships between activities.

In the area of resource determination, effort estimation has been extensively studied. However, it remains unclear which ML algorithm among those researched is the most effective, or which algorithms are best suited for specific situations. We agree with researchers that the application of a specific ML algorithm depends on the characteristics of that algorithm. Therefore, we encourage conducting experiments to identify the suitability of ML algorithms when addressing this challenge during implementation. However, resource determination also includes resource identification, an area that has not been sufficiently explored in the literature. The lack of studies on ML algorithms for this initial step of resource determination limits the potential for more efficient and scalable methods that leverage organizational data. In

our opinion, the resource identification aspect can be effectively addressed using ML in several ways. One option is to frame resource identification as a clustering problem, a type of problem for which ML techniques are well-suited. These techniques can uncover hidden patterns in large datasets without the need for predefined labels, allowing the system to suggest relevant resources for different activities based on their similarities. While this represents a complex challenge, we believe these methods could lead to effective solutions in this area.

Another potential direction for resource identification is to focus exclusively on identifying resource types, which appear to be a simpler task to the approach described earlier. The identification of resource type seems to be achievable using, e.g., ANN algorithms, DT, and CBR. For example, ANNs could help predict the resource type for a new activity by being trained on previous similar situations. DT could model decision-making rules for selecting appropriate resources based on defined attributes extracted from previous projects. CBR could analyze historical projects, where resource selection and identification were recorded, and compare current project conditions to past cases. These methods could lead to a resource recommendation solution that suggests the most relevant resources.

Additionally, we believe that the incorporation of NLP techniques could significantly enhance the resource determination subprocess. Since organizational context plays a critical role, NLP could help tailor resource identification to reflect organizational nuances in skillsets, resource types, and other requirements. A comprehensive resource identification solution should consider these contextual aspects, as industries and organizations often define their own nomenclature for resource management.

Moving to another subprocess, we also identified research gaps in duration estimation, particularly in determining which ML techniques are most suitable for forecasting activity durations. We believe that potential algorithms include ANN, DT, SVM, NB, RT, k-NN, ANFIS, and CBR. However, we have not identified any comparative studies that evaluate these ML algorithms for activity duration forecasting. Such studies would be valuable in uncovering the characteristics and behavior of a wide range of ML algorithms within the context of activity duration. These studies should utilize consistent datasets and should include benchmarking of ML methods against well-established statistical methods. Furthermore, the effectiveness of each technique should be assessed based on accuracy, robustness, data size, and noise tolerance. It is also important not to restrict ML experiments to the established input parameters of well-established activity duration methods, as this could hinder the discovery of more effective ML applications. Consequently, valuable ML applications could remain overlooked.

We believe that advancements in schedule optimization could lead to new methods for the overall schedule creation process. However, current studies on schedule optimization have been limited to resource optimization, omitting other optimization objectives. Specifically, a few reinforcement learning techniques have been applied to resource-constrained scheduling problems. To advance project scheduling, it is crucial to explore the application of reinforcement learning across various project scheduling problems, extending beyond resource optimization. We believe that a comprehensive study on the application of reinforcement learning to the most common scheduling problems could shed some light on the possibilities for implementing a universal ML-based schedule creation method. An end-to-end schedule creation solution would need to optimize the schedule based on the various objectives set by the project team.

Still, analyzing ML algorithms beyond reinforcement learning could provide new perspectives for schedule optimization subprocesses. While unsupervised and supervised learning seem to have more limited applicability to optimization problems compared to reinforcement learning, there is no conclusive evidence in the literature to support this due to the lack of studies encompassing the full spectrum of ML techniques in the schedule optimization area. Therefore, further methodological study could provide more light on this aspect.

Furthermore, we observed that the exploration of emerging ML techniques, such as graph neural networks (GNNs), has been limited across all project schedule creation processes. GNNs, which naturally handle graph-structured data, offer a promising new direction, particularly given the inherent graph-like structure of schedules. In our opinion, future studies on GNNs could provide a novel approach to project schedule creation and play a key role in schedule model optimization research. We believe that advancing research in this area could significantly contribute to the development of autonomous project scheduling.

Having highlighted key research gaps and opportunities in individual subprocesses, we now shift our focus to broader implications of how to implement ML in project scheduling processes in organizations. We recognize three potential paths for incorporating ML into project schedule creation. These paths offer a structured framework for enhancing schedule creation by leveraging the potential of ML:

- A. Adopting ML techniques to support existing project scheduling processes: This option involves integrating ML technique to enhance specific subprocesses, such as resource allocation or activity sequencing, without fully replacing human involvement.
- B. Automating one or more schedule creation subprocesses from an existing project scheduling method: This

approach is automating one or two subprocesses with computer-based systems with replacement of a human. For example, ML could be used to handle activity duration estimation or resource determination, while leaving other aspects to human decision-making.

- C. Developing a fully autonomous, end-to-end ML-driven project scheduling methodology: This model would enable the automation of the entire schedule creation process, allowing the project manager to focus on other aspects of project management.

Our findings show that ML techniques can enhance current project scheduling approaches, leading to improved results and increased planner performance. However, the lack of research on ML architectures for schedule creation and the need for further studies on almost every subprocess make it very challenging to develop a fully automated, end-to-end ML-driven project scheduling method. Given all considerations, we recommend pursuing option (B), as the current level of research supports this approach. Moreover, organizations typically possess sufficient data to support its potential implementation.

Given the current level of development, it is realistic to focus on solutions that automate specific subprocesses in widely used methods like CPM or PERT. For instance, ML techniques could enhance effort or duration estimation. We believe such automation would be feasible for organizations with sufficient historical data from past projects, as it is crucial for training ML models. In our view, ML-driven schedule creation enhancements are more likely to succeed in organizations with large datasets for specific project types. At the current level of research, we propose an incremental approach for the practical adoption of ML in project schedule creation: once ML has been successfully implemented in one subprocess, organizations can begin integrating it into the next one.

Importantly, the decision to support a particular project type with an ML scheduling method should be based on a well-prepared business case. The number of projects executed by an organization is an important factor in supporting the decision-making process, as these projects generate data for ML training. Therefore, it is crucial for organizations to create and continuously maintain datasets of past projects. A lack of sufficient data or poor-quality data from past projects will disqualify the use of ML in scheduling, potentially causing the opportunity for innovation to be missed or significantly delayed. Moreover, organizations should ensure they archive as much relevant information as possible, which can increase the predictability of ML algorithms used for project planning. For example, past project activity data should include detailed descriptions of activities and other related details, which can later be used for the automatic discovery of activity attributes throughout the entire schedule

creation process. Ultimately, the extent of data collected by organizations will be a decisive factor in the full adoption of ML techniques across all subprocesses.

The factors outlined above are crucial for adopting ML techniques to enhance existing project scheduling processes, as well as for fully automating one or more schedule creation subprocesses. However, implementing an autonomous project scheduling solution remains a significant challenge at the current stage of research on the use of ML in project schedule creation.

Approaching the topic from a radically different angle, both current research and our paper focus on leveraging well-established traditional ML techniques. These appear appropriate for enhancing schedule creation by automating certain aspects of the process and improving the schedule creation process incrementally. However, with the recent advancements in Large Language Models (LLMs), it is evident that the design of specialized architecture using unique ANNs has resulted in significant breakthroughs, particularly in generative applications. In the schedule creation area, the use of artificial neural networks has been primarily limited to specific tasks within the broader schedule creation process. The use of ANNs or deep learning models to address the entire schedule creation process holistically remains absent in the literature. In our opinion, given the current advancements, it should be feasible to investigate an overall architecture for an autonomous, end-to-end project schedule creation methodology, potentially inspired by LLM architecture and incorporating GNNs.

6 Conclusions

The expectation of the project management community is high regarding automatic and autonomous schedule creation. ML-based schedule creation should not only process a huge volume of data, but also generate a schedule that is complete, relevant, and optimized to project objectives. Considering the current research findings, it is plausible to apply ML to improve the project schedule creation. It remains uncertain whether, in the near future, the project schedule creation process will be a fully automated solution. Furthermore, it is still unknown if human involvement in the schedule creation process will be decreased to zero.

The key take away is that the schedule creation process has not received uniform attention across all subprocesses. While effort estimation has been extensively explored, other subprocesses still require further investigation. Notably, there is a significant gap in the literature regarding milestone definition. In terms of techniques, a wide range of ML techniques is utilized. Research in this field is dominated by studies from the software engineering domain.

According to the analyzed studies, ML techniques are generally as effective as, or superior to, traditional methods. However, the lack of publicly available, relevant datasets for developing and validating ML models remains the primary challenge. There is a noticeable absence of comprehensive papers outlining implementation strategies for integrating ML into project schedule creation.

However, we have elaborated on three potential development paths for integrating machine learning (ML) into the project scheduling process: (a) adopting ML techniques to support existing project scheduling processes; (b) automating one or more schedule creation subprocesses from an existing project planning method; (c) developing a fully autonomous, end-to-end ML-driven project scheduling methodology. We have indicated that option (b) is the most feasible path for implementation at the current stage.

In closing, it is important to acknowledge that our study primarily focuses on structuring emerging knowledge, providing suggestions for further research, and identifying the options for practical implementation that practitioners are seeking. It emphasizes the methodological and application aspects related to the business effectiveness of ML in project management processes.

At the same time, the ethical and social implications of this phenomenon must not be overlooked. The adoption of new automated scheduling methods is expected to significantly reshape the roles of schedulers, planners, and project managers. Furthermore, the emergence of new types of work driven by these technologies is likely to have a tangible impact on project practitioners.

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